Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/01/2006
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Project Title:	Searching for the Best Protein Crystals: Integration of Synchrotron-Based Crystal Quality Measurements and Structure Determination		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	68198-7696	**Congressional District:**	2
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	06/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Snell, E. (BAE Systems) Bellamy, H. (CAMD/LSU)		
Grant/Contract No.:	NAG8-1825		
Performance Goal No.:			
Performance Goal Text:			

The following specific hypotheses have been developed to test the general hypothesis.
1. Long range order can be measured by mosaicity to give a physical indication of domain structure within the crystal (tested in aim 1).

2. CCD detectors with high spatial resolution can be used for rapid, quantitative topography (tested in aim 2).

3. Topography is also a measure of long range order and can be correlated with mosaicity measurements (tested in aim 3).

4. Diffuse scatter and resolution enhancement can be used as a measure of the crystalline short range order (tested in

Task Description:

aim 4).

5. Combining the data from aims 1-4 will produce a description of the crystal that can be used to understand basic physical processes during microgravity crystal growth, during cryocooling and for structure determination (tested in aim 5).

The following specific aims will be performed.

1. Software enhancement.

2. Develop CCD methods of quantitative topography.

3. Correlate topography to mosaicity and intensity.

4. Measurement of diffuse scatter and diffraction resolution.

5. Use these methods for structure determination and to explore the mechanisms of microgravity crystal growth and cryocrystallography. The general hypothesis of this research is as follows. Microgravity growth improves the long and short range order in macromolecular crystals. How this happens is not well understood. A complete physical characterization of macromolecular crystals will improve our understanding and enable us to obtain the maximum benefit from microgravity crystal growth, from cryocrystallography and also to collect the highest resolution diffraction data for structure determination. A recent review of biotechnology research on the international space station (Space Studies Board, National Research Council, Washington, DC 2000) recommended that "...NASA must focus its support on programs that are developing technologically innovative equipment and engaging in the structure determination of crystals with important biological implications". We are engaged in both of these areas and have a unique, interdisciplinary team with expertise in each area of the work.

We are working to produce tools that will provide a complete X-ray based physical characterization of a crystal in a short period of time. In combination with this we will determine high resolution structures. We will be able to relate the physical nature of the crystals to the resulting structural results and thereby evaluate the apparatus, mission or technique producing the crystal. We will gain an understanding of how to make optimal use of the valuable and limited microgravity environment. We can answer questions that until now have eluded researchers such as what is the best growth technique, and what acceleration disturbances can crystals tolerate and still be better than ground-grown crystals. The tools are not only applicable to microgravity systems. We can look at techniques used in everyday structural crystallography – e.g. what is the best method of cryocooling? How does annealing work, when it does improve diffraction? By applying our instrumentation to crystals including MnSOD, human replication protein A, and human serum albumin we gain not only knowledge about the physical process but structural detail to further the understanding of life itself.

Specifically we will study crystals of several important macromolecules. We will use the techniques described here to identify the best crystals and then to achieve the most detailed structure possible. Aim 1, improvements to existing software, will extend our experiments to provide even more physical information from the data collected. We will extract precise structural information from our superfine fsliced data. Topography (aim 2 and 3) will relate the components of mosaicity to the crystal and produce a visual image of the components contributing to the Bragg reflections. Quantitative measurements of the contributions from the components will allow accurate modeling of the crystal and contribute to highly precise intensity determination. Diffuse scatter (aim 4) will allow us to look at differences in long and short range order improvement. Achieving these aims will give us the most advanced capability for assessing a crystal. We will use this in aim 5 to ask the questions described above. Practically, how applicable is this? Very. Modern third generation synchrotrons are increasingly using unfocused radiation. We have commonly seen reflection profile structure that most integration software would not evaluate accurately. Our software developments will overcome this and will further structural determination in general. The tools developed will become available to all microgravity scientists for application to their specific research and is therefore of value to the microgravity community as a whole.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The primary aim of this research project is to identify the optimum methods for protein crystal growth by quantitatively evaluating crystal quality and to collect high-resolution X-ray diffraction data for structure determination. This involves growing crystals of several proteins, developing methods to quantitatively assess crystal quality and subsequently collecting high quality diffraction data for subsequent structure determination at a synchrotron. Crystal quality is assessed by diffraction resolution, topography and mosaicity. These parameters are measured simultaneously using an innovative technique combining unfocused synchrotron X-ray beam, a CCD area detectors and superfine phi slicing. Research in FY06 was focused on the continued application of our developed CCD camera for topography measurements and publication during this no-cost extension period of the grant. Substantial progress was made and includes:
• CCD-based topography data was collected in the previous year on room temperature and cryo lysozyme crystals. In one particular data set, care was taken to take data on the exact same set of reflections at both temperatures. Several new topographic image analysis methods were introduced with this research. This work was presented at the 2005 International Union for Crystallography in Florence Italy and will be presenting it at the NSF NE EPSCoR Research Expo in March 2006 and was submitted and revised for publication in Acta Cryst. D. A manuscript concerning “Ripple”, the topography control software, was also submitted for simultaneous publication in J. Appl. Cryst.

• A proposal to the NSF was submitted on the modulated profilin:actin project. This work was also presented at the 2005 International Union for Crystallography in Florence Italy and warmly received.

• A “particle swarm” algorithm has been implemented for curve fitting the fine-sliced reflection profile data and an orthogonal projection algorithm was coded to extract the mosaic domains from the topographic image sequences. We plan to publish these methods this year in J. Appl. Cryst. and prepare a grant proposal to continue to fund this research.

• A cryocooling/reversibility study is planned with Drs. Doug Juers and Brian Matthews of the University of Oregon. Over the last year we have regrown the crystals and studied their annealing properties in preparation for a topography run at BioCARS at APS. The beamline at BioCARS has been down for renovation and we have plans to collect fine

	sliced reflection profiles and topography data there the summer of 2006. • Summary of FY06 publications submitted: 1. Jeffrey J. Lovelace, Cameron R. Murphy, Keith Brister, Reinhard Pahl and Gloria E. O. Borgstahl, “Tracking reflections through cryogenic cooling with topography” submitted to J. Appl. Cryst Nov. 22, 2005, revised Feb 6, 2006. 2. Jeffrey J. Lovelace and Gloria E. O. Borgstahl “Ripple: Ripple: A Program to Collect and Analyze Digital Topographic Sequences” submitted to J. Appl. Cryst. Nov. 22, 2005
Bibliography Type:	Description: (Last Updated: 08/10/2023)
Articles in Peer-reviewed Journals	Lovelace JJ, Murphy CR, Pahl R, Brister K, Borgstahl GEO. "Tracking reflections through cryogenic cooling with topography." Journal of Applied Crystallography. 2006 Jun;39(3):425-32. http://dx.doi.org/10.1107/S0021889806012763 , Jun-2006
Articles in Peer-reviewed Journals	Lovelace JJ, Borgstahl GEO. "Ripple: a program to collect and analyze digital topographic sequences." Journal of Applied Crystallography. 2006 Jun;39(3):466-7. http://dx.doi.org/10.1107/S0021889806014956 , Jun-2006
Articles in Peer-reviewed Journals	Lutz WE, Azadmanesh J, Lovelace JJ, Kolar C, Coates L, Weiss KL, Borgstahl GEO. "Perfect crystals: Microgravity capillary counterdiffusion crystallization of human manganese superoxide dismutase for neutron crystallography." npj Microgravity. 2023 Jun 3;9:39. https://doi.org/10.1038/s41526-023-00288-x ; PMID: 37270576; PMCID: PMC10238240 , Jun-2023
Presentation	Lovelace,J.J.; Murphy, C.R.; Pahl, R; Brister, K.; and Borgstahl, G. E. O. "Analyzing Mosaic Domain Changes Induced by Cryo-Cooling with Digital Topography" International Union for Crystallography, Florence, Italy Aug-2005
Presentation	Borgstahl,G.E.O.; Murphy, C.; Lovelace, J.; Narayan, K.; Svensson, C; Lindberg, L.; and Schutt, C.E. "Towards the Structure Determination of the Modulated Semicrystalline State of Profilin:Actin" International Union for Crystallography, Florence, Italy Aug-2005